Exhaust Manifold Assembly

Service and Diagnostic Manual

TITAN MOTORS MX-13 / Cummins X15 Engine Platforms

Part Number: EXM-4100

Revision: C

Date: January 2025

Classification: Dealer Restricted

Supersedes: All previous revisions

Section 1: General Description

The EXM-4100 Exhaust Manifold Assembly collects exhaust gases from the engine cylinders and directs them to the turbocharger inlet. The manifold operates continuously at temperatures exceeding 1,200 degrees Fahrenheit and must withstand severe thermal cycling over the life of the engine.

Section 2: Component Identification

The EXM-4100 consists of seven sub-component groups described in Table 2-1.

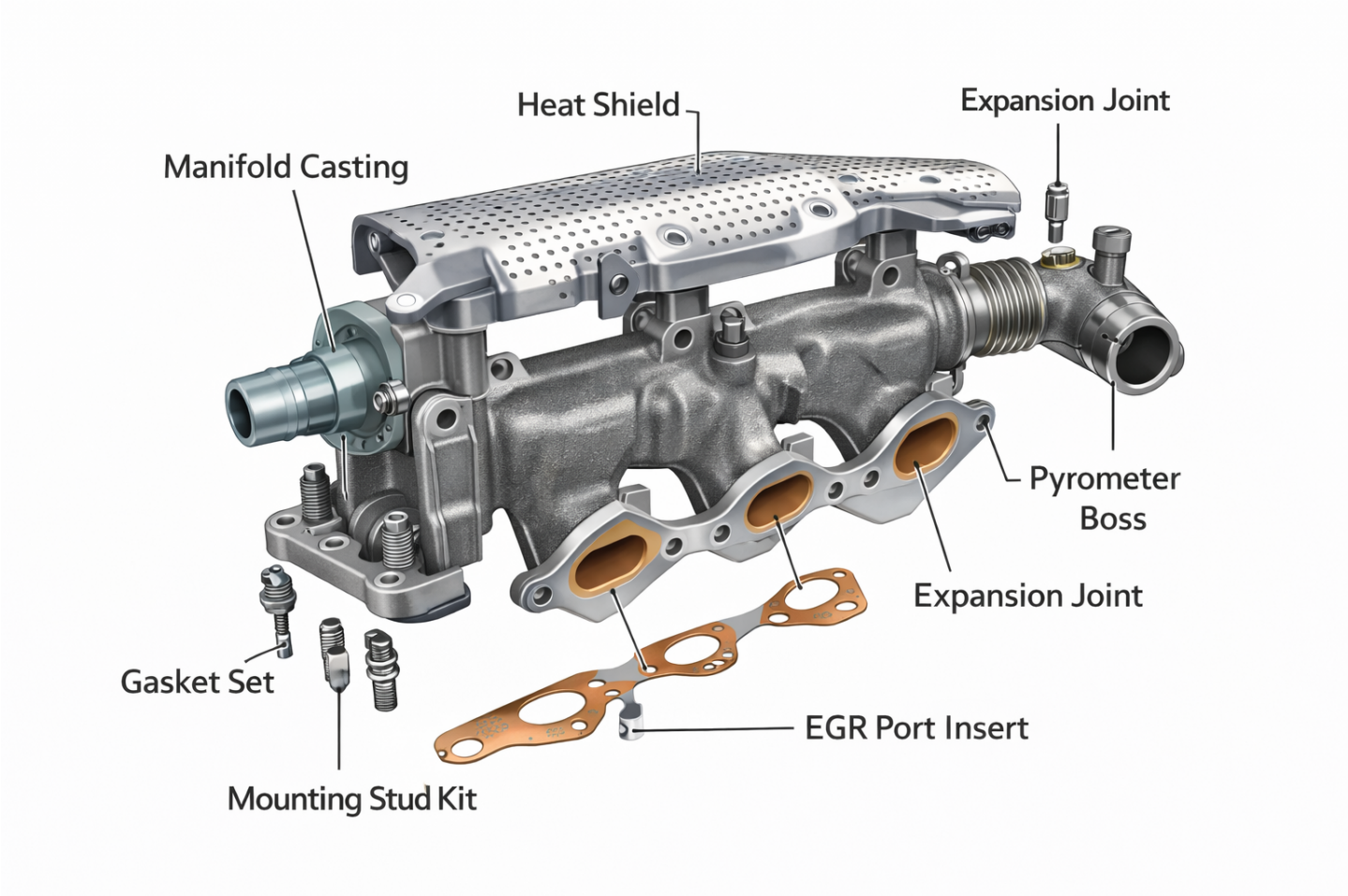

Figure 2-1: EXM-4100 Exhaust Manifold Assembly - Component Layout

Table 2-1: Sub-Component Identification

Item	Sub-Component	Description
1	Manifold Casting	High-chromium ductile iron (D5S Ni-Resist) primary structural element. Designed with controlled wall thickness variations to manage thermal stress distribution.
2	Gasket Set	Multi-layer steel (MLS) 3-layer gaskets providing cylinder-head-to-manifold sealing across six individual port interfaces. Must be replaced at every manifold installation.
3	EGR Port Insert	Stainless steel insert at the EGR pickup port. Directs a metered portion of exhaust gas to the EGR cooler circuit for NOx emissions control.
4	Heat Shield	Stamped stainless steel thermal barrier mounted on the cab-side of the manifold. Prevents radiant heat transfer to the engine wiring harness and cab-side components.
5	Mounting Stud Kit	Inconel 718 studs with controlled-stretch design. Secures the manifold to the cylinder head at 44 ft-lbs torque with anti-seize compound. Quantity: 12 per manifold.
6	Expansion Joint	Bellows-type stainless steel flex joint at the manifold-to-turbocharger interface. Absorbs angular misalignment and differential thermal expansion. Rated 500,000 cycles.
7	Pyrometer Boss	Welded M12 x 1.25 threaded boss for EGT sensor mounting. Positioned in the merged exhaust flow path downstream of the collector for accurate bulk gas measurement.

Section 3: Specifications

The following specifications apply to all EXM-4100 configurations.

Material	High-chromium ductile iron (D5S Ni-Resist)
Maximum Continuous Temperature	1,400 deg F
Peak Transient Temperature	1,600 deg F (< 30 seconds)
Mounting Stud Torque	44 ft-lbs (lubricated threads)
Mounting Stud Material	Inconel 718
Gasket Type	Multi-layer steel (MLS), 3-layer
Flex Joint Angular Range	+/- 2.5 degrees
EGT Sensor Boss Thread	M12 x 1.25
Assembly Weight	68 lbs
Flange Flatness Tolerance	0.004 inches max deviation

Section 4: Installation and Removal

Allow the engine to cool completely before attempting manifold removal. Remove the turbocharger assembly per the TC-5000 service manual. Disconnect the EGT sensor. Remove the heat shield mounting bolts. Remove mounting studs using a stud extractor - do not use pliers on stud threads. During installation, install new gaskets at every manifold installation, apply anti-seize to stud threads, and torque in the specified sequence to 44 ft-lbs.

Section 5: Theory of Operation

Thermal Management

During loaded highway operation, exhaust gas temperatures at the manifold ports typically range from 900 to 1,200 degrees F. During DPF regeneration, temperatures can spike to 1,400 degrees for 20 to 40 minutes. The manifold casting's thermal expansion during a cold-start-to-full-temperature cycle can exceed 0.040 inches along its length.

Expansion Joint Function

The expansion joint between the manifold outlet flange and the turbocharger inlet accommodates angular misalignment and absorbs differential thermal expansion. The bellows element has a rated life of 500,000 thermal cycles.

Section 6: Troubleshooting and Diagnostics

6.1 Exhaust Leak at Cylinder Port

Exhaust manifold leaks present with a characteristic ticking or tapping sound most prominent during cold start that diminishes as the engine reaches operating temperature. This occurs because thermal expansion closes the leak path as the manifold heats. The sound is rhythmic and synchronized with engine firing order. The most common cause is degradation of the gasket set at an individual cylinder port, typically at the number one or number six positions where thermal expansion displacement is greatest. Visible exhaust staining - a fan-shaped soot deposit on the manifold surface adjacent to the affected port - confirms the leak location.

If the gasket leak is not addressed within 20,000 to 30,000 miles, the escaping exhaust gas at 900 to 1,200 degrees F creates a localized hot spot on the manifold casting, generating thermal fatigue cracking in the casting wall. The mounting stud kit studs adjacent to the cracked area will develop elongation from uneven loading that cannot be corrected by re-torquing.

6.2 Exhaust Leak at Turbocharger Interface

A higher-pitched hissing at the manifold-to-turbo connection that does not diminish with warm-up indicates failure of the expansion joint bellows. When the expansion joint fatigues, it develops microscopic cracks in the bellows convolutions that initially allow only trace amounts of exhaust gas to escape. This seepage may produce a faint exhaust odor in the engine bay detectable only at idle, and is easily confused with residual exhaust from a parked-idle condition.

6.3 Phantom EGT Spikes

Intermittent spikes in exhaust gas temperature readings that do not correlate with load changes may indicate a cracked pyrometer boss rather than actual temperature excursions. The weld joint between the boss and the casting can develop hairline cracks from thermal cycling, allowing exhaust gas to impinge directly on the sensor element. These phantom EGT spikes can trigger unnecessary DPF regeneration events, increasing fuel consumption and accelerating DPF ash loading.

6.4 Heat Damage to Adjacent Components

A damaged or poorly mounted heat shield allows radiant heat from the manifold to reach components not designed for high-temperature exposure. The most common secondary damage is degradation of the engine wiring harness insulation on the intake side, resulting in intermittent electrical faults that appear unrelated to the exhaust system. Heat discoloration on the cab-side surface of the heat shield - a bronze or purple tint on what should be a silver metallic surface - indicates thermal over-exposure.

6.5 EGR Flow Restriction

Reduced EGR flow causing elevated NOx readings during emissions testing can result from carbon buildup in the EGR port insert. The insert's internal passage narrows over time as exhaust carbon deposits accumulate on the inner walls, restricting the metered flow to the EGR cooler. This restriction develops gradually and may not trigger an immediate fault code until the EGR flow deviation exceeds the ECM tolerance threshold.

Section 7: Preventive Maintenance

At each B-service, perform a cold-start listen test for exhaust tick sounds during the first 60 seconds of engine operation. At each C-service, inspect mounting stud torque - studs that have lost more than 15 percent of specified torque should be replaced. Inspect the expansion joint bellows for cracks or exhaust staining. At 300,000 miles, proactively replace the gasket set and mounting stud kit.